

AI-Powered Student Financial Document Management Platform

Mobile Application Implementation & Research Overview

EDUVOS | Faculty of Information Technology | ITDMA3-22 | 2026



Project Overview

RESEARCH CONTEXT

What We Are Researching

Students struggle to manage financial documents such as bursary letters, NSFAS correspondence and fee statements are scattered across emails and messaging apps.

Our study investigates how an AI-powered, cloud-native platform can be designed to help students securely organise, understand and act on these documents.

The research combines a quantitative student survey, qualitative interviews and technical architecture evaluation.



10+

Document Types

NSFAS, bursary, fee statements and more



100%

POPIA Aligned

Privacy-by-design architecture



3

Platforms

Mobile, Web and REST API

System Architecture

CLOUD-NATIVE LAYERS



Flutter Mobile App

Android & iOS



Next.js Web App

Browser dashboard



Spring Boot API

Auth · Logic · AI
gateway



AI Processing

LLM · RAG ·
Embeddings



AWS S3 + PostgreSQL

Storage · Metadata ·
Audit

All client apps communicate exclusively through the Spring Boot API never directly with storage or AI services. This enforces access control and audit logging at every step.

Mobile Application: Why Flutter?

TECHNOLOGY CHOICE



Single Codebase

One codebase compiles to both Android and iOS. This halves development effort and keeps features consistent across platforms.



Rich Widget Library

Flutter's widget system makes it straightforward to build document cards, chat-style AI response threads and notification banners without third-party UI frameworks.



Compiled Performance

Flutter compiles to native ARM code rather than interpreting at runtime. This means smooth scrolling, fast document rendering and responsive AI interactions.



REST API Integration

Flutter's http and dio packages integrate cleanly with the Spring Boot backend. JWT token management and file upload handling are well supported.

Mobile App: Key Screens & Features

USER FEATURES



Document Upload

- Camera capture or file picker
- Supports PDF, JPG, PNG
- Progress indicator during upload
- Auto-classification on arrival



AI Smart Search

- Ask questions in plain language
- Answers grounded in your own docs
- Source citations shown per answer
- Filters by document type or date



Document Summary

- Plain-language summary per doc
- Key dates and amounts extracted
- Conditions and obligations flagged
- One-tap access to full document



Smart Notifications

- Push alerts for upcoming deadlines
- Conditions requiring action flagged
- Renewal reminders for bursaries
- Configurable per document type



Secure Access

- Biometric and PIN login
- OAuth 2.0 token-based auth
- Session timeout on inactivity
- Encrypted local document cache



AI Chat Interface

- Chat-style question and answer
- Multi-turn conversation support
- RAG retrieval per message
- Responses cite document source

Document Processing Pipeline

BACKEND FLOW

01

Upload

Student selects document in Flutter app. Backend validates session, creates metadata record and returns pre-signed S3 URL.

02

OCR & Parse

Text-native PDFs are parsed directly. Scanned or photo documents go through OCR. Confidence scores are recorded.

03

Chunking

Extracted text is split into overlapping chunks tagged with doc ID, page and section. Chunk size balances context and retrieval accuracy.

04

Embedding

Each chunk is converted into a dense vector by the embedding model and stored in the vector database with access metadata.

05

AI Analysis

LLM generates a plain-language summary, document classification and extracted entities (amounts, dates, conditions).

AI Question Answering: How RAG Works

RETRIEVAL-AUGMENTED GENERATION

Student question (Flutter chat):
"What are the conditions of my NSFAS award?"

Answer shown in chat with source reference:

"You must maintain 60% pass rate. Source: NSFAS_Award_2026.pdf, p.3"

1

Query Embedded

The question is converted into a vector using the same embedding model as the documents.

2

Permission Check

Backend confirms the student's session and restricts retrieval to their own documents only.

3

Vector Search

The query vector is compared against stored chunk embeddings. Top-k most relevant passages retrieved.

4

Prompt Built

Retrieved passages are inserted into the LLM prompt as context alongside the student's question.

5

Answer Generated

LLM produces a response grounded in the retrieved passages, with source citations attached.

Security & Privacy Architecture

Our core ethical commitment: student financial data is private and must stay that way.



OAuth 2.0 + OpenID Connect

Keycloak or AWS Cognito manages student login, token issuance, role assignment and MFA. JWT tokens validated on every API request.



Student-Level Access Control

RBAC defines system roles. Document-level metadata filters ensure no student can access another student's files, even accidentally.



Encryption Everywhere

S3 server-side encryption for stored documents. HTTPS/TLS for all traffic. Secrets stored in AWS Secrets Manager, never in code.



Immutable Audit Logs

Every upload, view, AI query and deletion is logged. Logs are immutable and retained per institutional policy. Supports POPIA compliance.



POPIA Compliance

Data minimisation, purpose limitation, consent management and deletion workflows are embedded in the architecture, not added as an afterthought.



AI Safety Controls

RAG retrieval filtered by user permissions. AI outputs include source citations. Prompt injection risks mitigated through input sanitisation and separation of system and user content.

Mobile Authentication Flow

OAUTH 2.0 + OPENID
CONNECT

1

Student Opens App

Flutter app checks for a valid cached JWT token. If expired or absent, redirects to login.

2

Identity Provider

Keycloak or AWS Cognito presents the login screen. Supports email/password and optionally social login.

3

MFA Challenge

If MFA is enabled, student completes one-time password or authenticator app verification.

4

Token Issued

Identity provider returns access token and refresh token. Flutter stores these securely using flutter_secure_storage.

5

API Requests

Every API call includes the access token in the Authorization header. Spring Boot validates it on each request.

6

Token Refresh

Before token expiry, Flutter automatically exchanges the refresh token for a new access token in the background.

Technology Stack Summary

FULL STACK

Technology	Role	What It Does in Our Platform
Flutter	Mobile Application	Android & iOS from a single Dart codebase. Handles upload, AI chat, push notifications and secure token storage.
Next.js	Web Application	React-based dashboard with server-side rendering. Supports bulk upload, analytics and account management.
Spring Boot	Backend REST API	Manages authentication, business logic, document workflow orchestration and AI service integration.
PostgreSQL	Relational Database	Stores document metadata, user records, workflow states, permissions and audit log entries.
AWS S3	Object Storage	Encrypted, scalable storage for raw financial documents. Pre-signed URLs control upload and download access.
Pinecone / ChromaDB	Vector Database	Stores document chunk embeddings. Supports permission-filtered semantic search for RAG retrieval.
Keycloak / AWS Cognito	Identity Provider	OAuth 2.0 and OpenID Connect. Manages student accounts, role assignment, token issuance and MFA.
LangChain + LLM API	AI Orchestration	Coordinates chunk retrieval, prompt construction, LLM API calls and response formatting with citations.

Research Methodology

DESIGN SCIENCE RESEARCH



Quantitative Survey

100–
150

Online survey to students receiving financial aid
Measures document management pain points
Likert-scale items on AI feature preferences
Identifies privacy and security concerns
Analysed using descriptive statistics



Qualitative Interviews

12–18

Semi-structured interviews via Teams/Zoom
Explores trust, workflow needs and expectations
Thematic analysis using Braun and Clarke (2006)
Purposive sampling across aid types
Transcribed and coded for themes



Technical Evaluation

6

Architecture evaluated against defined criteria
Security, scalability, POPIA compliance checked
Partial prototype where feasible
AI retrieval quality assessed through human review
Evaluation criteria from cloud-native frameworks



Building Responsibly.

Student financial data is sensitive. Our platform is designed with privacy-first architecture, POPIA compliance and AI transparency as non-negotiable foundations.

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